

**DEPARTMENT OF DATA SCIENCE**  
**II CSD REAL TIME RESEARCH PROJECT**

**A.Y. 2025-2026**

**Date: 11/02/2026**

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| <b>Domain / Areas</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Data Science and Artificial Intelligence (Computer Vision)                                                                                                                                                                                             |             |
| <b>Title of the Project</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | AI POWERED POTHOLE DETECTION                                                                                                                                                                                                                           |             |
| <b>Team Leader Name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | V.V.MITHILESH (24R21A67C8)                                                                                                                                                                                                                             |             |
| <b>Team Members</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | <b>Roll No</b>                                                                                                                                                                                                                                         | <b>Name</b> |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 24R21A67C6                                                                                                                                                                                                                                             | SANKEERTHYA |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 24R21A67B9                                                                                                                                                                                                                                             | VIKRAM      |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 24R21A67C0                                                                                                                                                                                                                                             | VINEETH     |
| <b>Guide Name</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | MRS.DIVYA                                                                                                                                                                                                                                              |             |
| <b>Guide Signature</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                        |             |
| <p><b><u>A BSTRACT</u></b> : Road infrastructure plays a crucial role in economic development and public safety. However, potholes and road surface damages are major challenges in urban and rural transportation systems. These defects often lead to traffic accidents, vehicle damage, increased fuel consumption, and traffic congestion. Traditional road inspection methods are manual, time-consuming, costly, and inefficient. To address this issue, this project proposes an AI-based Pothole Detection System using Deep Learning techniques, specifically Convolutional Neural Networks (CNN) and YOLO (You Only Look Once) object detection models.</p> <p><b>Keywords</b> :Pothole Detection, Deep Learning, Convolutional Neural Network ( CNN), YOLO (You Only Look Once),Computer Vision ,Image Processing, Road Damage Detection</p> |                                                                                                                                                                                                                                                        |             |
| <b>Software Needs</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Programming Language: Python, IDE: VS Code / PyCharm / Jupyter Notebook<br>Deep Learning Framework: TensorFlow or PyTorch , Object Detection Model:<br>YOLOv5 / YOLOv8 Image Processing: OpenCV ,Libraries: NumPy, Pandas,<br>Matplotlib, Scikit-learn |             |

**Signature of the Coordinator**

**Signature of the HOD(CSE-DS)**